

Problem Statement and Goals

Natural Control

Group 7

Aaron Loh

Three team member names pending confirmation

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Review draft – September 23, 2026. Prepared from Aaron Loh’s planning interview. Team-wide ratification, remaining names, approvals, and reflections are pending. Proposed targets are not achieved results. This document is not ready for submission.

Table 1: Revision history.

Date	Revision	Change
2026-09-23	Review draft	AI-assisted draft from Aaron Loh’s decisions; whole-team review pending.

Table 1 records this draft’s origin. Natural Control is a working name. The accompanying Development Plan contains the proof of concept (POC) plan and proposed team charter.

1 Problem Statement

Blind computer users can encounter barriers when completing tasks across websites, applications, and connected services. The project addresses independent task completion and the ability to remain informed and in control when delegating digital work.

1.1 Problem

Completing a digital task can require navigating unfamiliar interfaces, tracking changes, and recovering from controls or content that existing access tools do not expose effectively. Screen readers already provide selective navigation; users do not simply listen to every item on a page. WebAIM’s self-selected survey identifies barriers including unclear controls and unexpected changes, but does not establish their prevalence among our prospective participants [3].

Delegation to an artificial intelligence (AI) agent introduces another problem: the user needs to understand what is happening, make meaningful choices, and correct mistakes without visual monitoring. Research on Morae examines this issue by involving blind and low-vision users in decisions during agent execution [1]. These sources motivate the project; interviews must still establish participants’ actual priorities.

Natural Control aims to make general digital task delegation usable through conversation while preserving user agency. Success means a person can complete and verify useful tasks independently after initial setup, not that they abandon their screen reader. Representative tasks span connected accounts, unfamiliar websites, and local computer applications. They are evaluation cases, not a product allowlist or fixed scripts.

1.2 Inputs and Outputs

The intended inputs are spoken goals, preferences, clarification answers, corrections, approvals, and stop requests. Users may also supply keyboard or text input. With appropriate access, the system receives relevant page, application, file, and connected-account state needed for the requested task. Initial setup supplies permitted accounts, tools, and provider credentials; credentials are not intended for spoken entry or public logs.

The intended outputs are actions on permitted services or the computer; concise spoken progress and results; requests for clarification or approval; and an accessible, reviewable task history. Completion feedback must distinguish verified results, partial work, and unresolved failures. An acknowledgment of a stop request is not a claim that execution has already stopped or that prior actions were undone.

1.3 Stakeholders

Primary users.

Blind people with screen-reader experience who want to delegate digital tasks while retaining control. Initial evaluation is with English-speaking Mac users; this is a development boundary, not a claim that these users represent all blind people.

Research collaborators and accessibility organizations.

Potential participants and advisers can identify barriers and evaluate interaction choices. A team connection may facilitate a CNIB introduction; no partnership, participation, or endorsement is confirmed.

Setup helpers and open-source contributors.

Trusted helpers may assist with initial installation and account setup. Maintainers need understandable code, documented limitations, and a reproducible default configuration.

Course team and assessors.

Four student contributors, the teaching assistant (TA), and instructor need a feasible project and evidence that the team understands its work. People receiving messages or invitations are also affected by the system's actions, which makes accurate user authorization important.

1.4 Environment and Boundaries

The initial evaluated environment is macOS with Chrome, English speech, a microphone, audio output, internet access, and existing accessibility tools such as VoiceOver and keyboard navigation. A web voice interface is planned; its location in a browser does not restrict the agent to web tasks. Cloud processing is permitted, and users provide their own account connections and provider credentials.

A local component is necessary for local Mac tasks. An always-available Mac or Mac mini is one deployment option; hosted execution is another for tasks that do not require a local device. A hosted service cannot operate an offline Mac. Independent everyday use is the initial goal; assisted setup is acceptable. Telephone access will be compared with on-Mac voice, with support for both channels treated as an extension.

Unrestricted success on every website or application, guaranteed reversibility, universal provider compatibility, and replacement of all screen-reader functions are not promised. The project will build on an existing agent rather than train a new foundation model. User needs and provider constraints may change the evaluated scope, with changes recorded rather than hidden.

2 Goals

The following four goals describe benefits to users. Numerical targets are provisional design commitments accepted for this draft, not measured results. A pilot will inform revisions before the main evaluation; changes will be justified and retained in the revision history. Detailed task selection and measurement procedures belong in the Development Plan and later verification and validation plan.

- G1. **Complete useful digital tasks independently.** After assisted setup, users should complete at least 80% of a declared, representative task set correctly without sighted assistance. The set will cover connected-account, browser, and local-Mac tasks. Report results per category and overall, including failed attempts. This target applies to the evaluated tasks, not arbitrary computer use.
- G2. **Retain control over delegated work.** Users should be able to stop work and decide whether consequential actions proceed. In the evaluation, explicit stop requests should be acknowledged within two seconds, and every tested consequential action must require explicit approval. Verified stopping is separately checked: no new action may be issued after the system confirms execution stopped.
- G3. **Understand and correct the assistant without visual help.** In each evaluated scenario, check whether the participant can identify the actual outcome or blocker and, in a seeded-misunderstanding scenario, correct the request without sighted assistance. Report successful cases and failure reasons. Brief feedback and access to details should support awareness without requiring constant narration.
- G4. **Make the system usable beyond the development team.** Release an open-source implementation with one documented default configuration, user-provided credentials, and extension points. Verify that a person outside the implementation team can follow the setup guide on a clean supported environment, with assistance documented, and run the reference tasks. Stars or downloads alone are not evidence of usefulness.

3 Stretch Goals

The leading extension is a second voice channel: let a user access the same task context through telephone and on-Mac voice. Later possibilities include messaging continuity and evaluated Safari support. These must not displace a reliable initial experience or be presented as completed work.

4 Custom Documents (CDs)

Two custom documents are proposed under the course outline [2]. **Instructor approval is pending.**

- **Fall: User/Stakeholder Interview Report.** Record recruitment, participant needs, methods, findings, and resulting design decisions. Separate direct evidence from team hypotheses.
- **Winter: Usability Report.** Evaluate independent completion, awareness, correction, and user control against participants' usual workflows. Report limitations and unsuccessful cases as well as benefits.

Appendix A: Generative AI Use Disclosure

Tool and use. Aaron Loh used OpenAI Codex with GPT-6 Astra, primarily at Medium reasoning and occasionally High (settings reported by Aaron). Codex helped extract questions from course requirements, challenge assumptions, research options, and develop ideas through discussion. It also assisted with drafting and revising text throughout this document, organizing references, and LaTeX/PDF formatting. Aaron made the judgment calls and reviewed the work. These are working drafts for further team revision, not evidence of team-wide agreement or completed participant research.

Verification and responsibility. The assistant checked cited sources, course requirements, document consistency, and PDF rendering. These checks do not establish that the proposed system works. Team review, ratification, and personal reflections remain pending. Before submission, the team will verify the final content and update this disclosure to reflect subsequent revisions and any additional AI use. The team remains responsible for the submitted work.

Appendix B: Reflection

These prompts require the team's own account of producing the deliverable. They have deliberately not been answered on behalf of absent contributors.

1. **What went well? Pending team input:** Describe actual collaboration and what helped produce this deliverable.
2. **What pain points arose and how were they resolved? Pending team input:** Supply actual experiences and actions, not hypothetical lessons.
3. **How did the team adjust scope? Pending team input:** Reflect on the scope decision. The planning record establishes a general-agent interface with bounded evaluation, but does not establish every member's reasoning or agreement.

References

- [1] Morae. Morae: Proactively pausing ui agents for user choices. <https://arxiv.org/abs/2508.21456>, 2025. Research paper, arXiv:2508.21456. Accessed September 23, 2026.
- [2] Spencer Smith. Sfwreng 4g06 course outline, 2026. August 31, 2026. User-supplied course document; local source indexed in refs/course/README.md.
- [3] WebAIM. Screen reader user survey #10 results. <https://webaim.org/projects/screenreadersurvey10/>, 2024. Accessed September 23, 2026. Self-selected survey, not population prevalence.